

Document Summary TASK-DC71CB09

Analysis & Generated Response

Certainly! Below is a complete Python code snippet for a simple calculator that can perform basic arithmetic operations such as addition, subtraction, multiplication, and division. This code includes comments to explain each step of the process and usage instructions at the end.

```
```python
```

```
Import necessary libraries
```

```
import sys
```

```
def add(a, b):
```

```
 """
```

```
 Add two numbers.
```

```
 Parameters:
```

```
 a (float): The first number.
```

```
 b (float): The second number.
```

```
 Returns:
```

```
 float: The sum of a and b.
```

```
 """
```

```
 return a + b
```

```
def subtract(a, b):
```

```
 """
```

```
 Subtract the second number from the first.
```

```
 Parameters:
```

```
 a (float): The minuend.
```

```
 b (float): The subtrahend.
```

```
 Returns:
```

```
 float: The difference between a and b.
```

```
 """
```

```
 return a - b
```

```
def multiply(a, b):
```

```
 """
```

Multiply two numbers.

Parameters:

a (float): The first number.

b (float): The second number.

Returns:

float: The product of a and b.

```
"""
```

```
return a * b
```

```
def divide(a, b):
```

```
"""
```

Divide the first number by the second.

Parameters:

a (float): The dividend.

b (float): The divisor.

Returns:

float: The quotient of a divided by b. If b is zero, returns None.

```
"""
```

```
if b == 0:
```

```
 return None
```

```
 return a / b
```

```
def main():
```

```
 # Print welcome message
```

```
 print("Welcome to the Simple Calculator!")
```

```
 while True:
```

```
 try:
```

```
 # Prompt user for input
```

```
 operation = input("Enter an operation (+, -, *, /) or 'exit' to quit: ")
```

```
 if operation.lower() == 'exit':
```

```
 break
```

```
 num1 = float(input("Enter the first number: "))
```

```
 num2 = float(input("Enter the second number: "))
```

```
Perform the selected operation

if operation == '+':
 result = add(num1, num2)
elif operation == '-':
 result = subtract(num1, num2)
elif operation == '*':
 result = multiply(num1, num2)
elif operation == '/':
 result = divide(num1, num2)
else:
 print("Invalid operation. Please try again.")
 continue

#
```

## Grounding & Evidence Sources

- **sacnned pdf 2 1.pdf** (Referenced): Sample Scanned Copy: Academic Year : 2012■2013  
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